An Exact Computer-Assisted Lower Bound for Packing 17 Unit Squares in a Square

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https://github.com/Mira-acc/17squares

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Abstract

Let $s(n)$ denote the least side length of a square that contains n pairwise interior-disjoint unit squares, with arbitrary orientations. We prove by an exact computer-assisted argument that

$$s(17) > \frac{4\,468\,292}{1\,000\,000} = 4.468292.$$

The previous exact computer-assisted bound was $s(17) > 4.456575$. Our proof gives sixteen rational points in the square of side 4.468292 and certifies that every contained unit square, at every orientation, contains one of these points strictly in its interior. Besides direct point witnesses, the certificate uses a strict triangle-piercing lemma to cover center regions independently of orientation. The result then follows by the pigeonhole principle. The universal geometric assertion is checked by a 122,626,747-node recursive bisection of the three-dimensional pose space. All decisive comparisons are exact integer inequalities. A fast C++ checker and separately implemented arbitrary-precision C++ and Python checkers validate the same certificate.

1 Introduction

For $L > 0$, write

$$Q_L = [0, L]^2.$$

Let $s(n)$ be the minimum L for which n pairwise interior-disjoint unit squares can be placed in Q_L, with arbitrary translations and rotations. The minimum is attained by the standard compactness argument: centers and orientations range over a compact parameter space, and the containment and nonoverlap constraints are closed. This classical packing problem has a long history; Friedman's dynamic survey collects the principal constructions and lower-bound methods[1]. For $n = 17$, the survey records the lower bound

$$s(17) \geq \frac{40\sqrt{2} + 19}{17} = 4.445208382054341\ldots,$$

attributed there to Trevor Green. The best known construction is due to John Bidwell and has side approximately 4.67553009360455[1, 3].

The first exact certificate in this line proved $s(17) > 4.450837$. Fort then optimized the rational point set within the same point-witness architecture and proved $s(17) > 4.456575$[4]. The certificate below changes the point set again and adds an orientation-independent triangle witness.

Our result is the following.

Table 1: The sixteen rational witness points.

i	x_i	y_i	i	x_i	y_i
1	0.9633740	0.9849850	9	0.9172158	2.6508741
2	1.6317845	0.9829523	10	1.4694893	2.6410332
3	2.5287879	0.9446006	11	2.4689011	2.6753176
4	3.4682923	0.9443212	12	3.4682923	2.7101964
5	0.9999997	1.7580956	13	0.9999997	3.5239708
6	1.9993909	1.7929744	14	1.9395041	3.5236914
7	2.9988027	1.8272588	15	2.8365087	3.4853407
8	3.5451063	1.8219620	16	3.4687946	3.4687946

Theorem 1. *No packing of seventeen unit squares exists in Q_L for*

$$L = \frac{4\,468\,292}{1\,000\,000}.$$

Consequently,

$$s(17) > 4.468292.$$

The improvement over the preceding exact bound is 0.011717. The proof uses the unavoidable-point method associated with Stromquist and developed in this setting by Friedman[1, 2]. The new contribution is an explicit rational point set, the strict triangle-piercing leaf described below, and an exact certificate covering the full continuum of square positions and orientations.

2 Strictly unavoidable points

Definition 2. A finite set $P \subseteq Q_L$ is *strictly unavoidable for unit squares* if every unit square $U \subseteq Q_L$ satisfies

$$P \cap \operatorname{int}(U) \neq \varnothing.$$

Lemma 3 (Pigeonhole lemma)**.** *If Q_L contains a strictly unavoidable set of $n-1$ points, then no packing of n pairwise interior-disjoint unit squares exists in Q_L.*

Proof. Suppose $U_1, \ldots, U_n$ were such a packing. For each i, choose $p_i \in P \cap \operatorname{int}(U_i)$. If $p_i = p_j$ for $i \neq j$, then that point lies in both square interiors, contradicting interior-disjointness. Thus the map $i \mapsto p_i$ is an injection from an n-element set into an $(n-1)$-element set, which is impossible. □

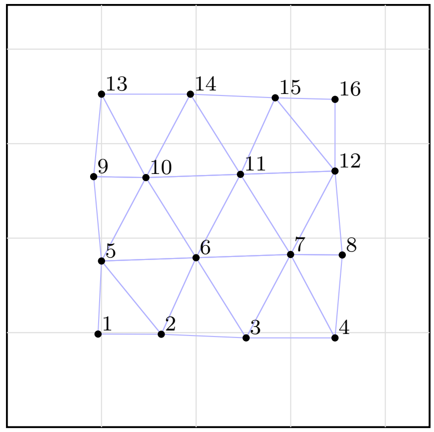
The use of interiors is important. Boundary witnesses could be shared by several touching squares and would not imply the injection in Lemma 3.

For Theorem 1, the set P consists of the sixteen rational points in Table 1. Every displayed decimal is exact, with denominator 10^7.

3 Parameterizing a unit square

A unit square is unchanged by a rotation through $\pi/2$. We may therefore represent every orientation by an angle

$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}, \qquad t = \tan\theta \in [-1, 1].$$



$L = 4.468292$

Figure 1: The witness set and the eighteen certified triangles. The proof uses the exact rational coordinates in Table 1; blue edges indicate the orientation-independent triangle witnesses.

Let (x, y) be the center of the square, and let $p = (p_x, p_y)$ be a fixed point. Put

$$\Delta x = p_x - x, \qquad \Delta y = p_y - y, \qquad r = \sqrt{1+t^2}.$$

In coordinates aligned with the square, the displacement from the center to p is

$$\left(\frac{\Delta x + t\Delta y}{r}, \frac{-t\Delta x + \Delta y}{r} \right).$$

Consequently, p lies strictly inside the square exactly when

$$4(\Delta x + t\Delta y)^2 < 1 + t^2, \tag{1}$$

$$4(-t\Delta x + \Delta y)^2 < 1 + t^2. \tag{2}$$

The axis-parallel half-extent of a unit square with parameter t is

$$h(t) = \frac{1 + |t|}{2\sqrt{1+t^2}}.$$

Thus containment in Q_L requires

$$h(t) \le x, y \le L - h(t).$$

The function $u \mapsto (1+u)/(2\sqrt{1+u^2})$ is nondecreasing on $[0, 1]$. This monotonicity supplies a simple exact test for pose boxes that cannot contain any feasible square.

4 The subdivision certificate

The pose space is parameterized by (x, y, t). The checker begins with the rational box

$$\mathcal{B}_0 = \left[\frac{1}{2}, L - \frac{1}{2}\right]^2 \times [-1, 1],$$

which contains every feasible pose. It recursively bisects boxes in one of the three coordinates. Each node is encoded by one byte:

Byte	Meaning
0	the current box contains no feasible square pose;
$1, \ldots, 16$	the indicated point lies strictly inside every pose in the box;
$17, 18, 19$	bisect the box in x, y, or t, respectively.
$20, \ldots, 37$	the indicated triangle pierces every pose in the box.

The verifier distrusts every leaf claim and recomputes it from the box endpoints.

4.1 Infeasible leaves

For a box

$$\mathcal{B} = [x_0, x_1] \times [y_0, y_1] \times [t_0, t_1],$$

let

$$u = \min\{|t| : t \in [t_0, t_1]\}.$$

Since $h(t) \geq h(u)$ throughout the box, the box is infeasible if any one of

$$x_1 < h(u), \quad L - x_0 < h(u), \quad y_1 < h(u), \quad L - y_0 < h(u)$$

holds. The checker clears denominators and squares nonnegative quantities, so this test is performed using integers only.

4.2 Point-witness leaves

Fix a witness point p. Exact interval arithmetic over $\mathcal{B}$ gives intervals containing

$$A = \Delta x + t\Delta y, \qquad B = -t\Delta x + \Delta y.$$

Write M_A and M_B for the maximum absolute endpoint of these intervals. Because $1 + t^2 \geq 1 + u^2$ throughout the box, the two strict inequalities

$$4M_A^2 < 1 + u^2, \qquad 4M_B^2 < 1 + u^2$$

prove simultaneously that p is strictly inside every square represented by that box. Again, the implementation stores rational endpoints on fixed grids and clears every denominator before comparison.

The interval enclosure for the bilinear term $t\Delta y$ is obtained by taking the minimum and maximum of its four corner products; the same is done for $(-t)\Delta x$. No floating-point approximation enters a leaf decision.

4.3 Triangle-witness leaves

Some center boxes can be closed without subdividing the orientation interval. The certificate fixes eighteen triangles whose vertices belong to the witness set and applies the following lemma.

Lemma 4 (Strict triangle piercing)**.** *Let A, B, C be three points whose pairwise distances are strictly less than one. Every unit square whose center lies strictly inside triangle ABC contains at least one of A, B, C strictly in its interior.*

Proof. Translate the square center to the origin and rotate the plane so the square is $[-1/2, 1/2]^2$. Its diagonals divide the plane into right, top, left, and bottom closed cones. Because the origin lies strictly inside triangle ABC, the three vertices cannot all lie in a closed half-plane bounded by either diagonal. Among the four cones this forces two vertices into opposite cones. If neither vertex were in the square interior, vertices in the right and left cones would have x-coordinates differing by at least one; vertices in the top and bottom cones would have y-coordinates differing by at least one. In either case their Euclidean distance would be at least one, contradicting the strict side-length hypothesis. □

For a triangle leaf, the checker first verifies the three strict inequalities

$$\|A-B\|_2^2 < 1, \qquad \|B-C\|_2^2 < 1, \qquad \|C-A\|_2^2 < 1.$$

It orients the triangle counterclockwise and checks that all four corners of the center rectangle lie strictly to the left of each directed edge. Each edge test is affine in the center coordinates, so positivity at the four corners proves positivity throughout the rectangle. Lemma 4 then covers every orientation in the box. The checker clears denominators and performs these tests with integers only.

4.4 Completeness of the tree

The checker traverses the tree from $\mathcal{B}_0$, applying every split exactly. It rejects an early end of file, an unknown opcode, a false leaf claim, or any trailing byte. Finally it checks the full-binary-tree identity

$$\#\text{leaves} = \#\text{split nodes} + 1.$$

It follows inductively that the accepted leaves form a cover of the complete root box. Every feasible pose is therefore covered by either a point-witness leaf or a triangle-witness leaf. In both cases at least one of the sixteen points lies strictly inside the square, so the point set is strictly unavoidable.

Lemma 3 now rules out a packing in Q_L, and therefore in every smaller container as well. Thus $s(17) \geq L$. The minimum defining $s(17)$ is attained by compactness, so equality would give a packing in Q_L. Hence $s(17) > L$, proving Theorem 1.

5 Exact arithmetic and reproducibility

The certificate uses the scales

$$S_t = 2^{40}, \qquad S_x = 10^6\, 2^{40}, \qquad D_p = 10^7.$$

The center and slope endpoints are stored as integers on these fixed rational grids, while the witness points have denominator D_p. The accepted tree has

Total nodes	122,626,747
Split nodes	61,313,373
Point-witness leaves	33,762,069
Triangle-witness leaves	483,875
Infeasible leaves	27,067,430
Maximum depth	74

The certificate consists of one byte per node. Its raw SHA-256 digest is

`2838f315302d67da131745925e9ec7dd2a602bb299d1335ce25e4e13a7b7b6d2.`

The raw tree is 122,626,747 bytes, above GitHub's ordinary per-file limit. The repository stores a deterministic 374,096-byte XZ archive with SHA-256

`a349b81e630ccf7292ae0afe6ed954591f7e88fe6289847f67883204a7ed60ac.`

The repository provides three checkers:

1. a fast C++ checker using bounded wide integers and the same arithmetic header as the generator;
2. a separately implemented C++ checker using arbitrary-precision integers;
3. a separately implemented Python checker using arbitrary-precision integers.

The generator and its search heuristics are not part of the trusted proof core: each checker reads the certificate as untrusted input and recomputes every leaf claim. The two arbitrary-precision checkers do not reuse the generator's arithmetic routines, and so do not rely on a fixed-width overflow analysis. From the repository root, the command

```
bash ./verify_all.sh
```

regenerates the deterministic certificate, checks its hash, compares it byte-for-byte with the archive, and executes the three verifiers. The proof package is available at `https://github.com/Mira-acc/17squares`.

6 Discussion

This result is a numerical improvement and a modest extension of the certificate language. The mathematical proof has two layers: the short unavoidable-point argument in Section 2 and an exact finite certificate for its universal geometric premise. Triangle leaves account for 483,875 pose boxes and discharge their complete orientation intervals at once. The point search and the certificate-generation heuristic may be useful for finding the proof, but neither must be trusted to verify it.

The current gap remains substantial:

$$4.468292 < s(17) \leq 4.67553009360455\ldots.$$

The certificate does not imply that the sixteen-point set is optimal, nor does it provide evidence that Bidwell's construction is globally optimal. Natural next steps are external reproduction, formal verification of the small trusted kernel, optimization of unavoidable point sets, and construction searches informed by the least-covered poses.

Disclosure. The computation and initial exposition were developed with assistance from OpenAI's GPT-5.6 Pro under human direction. The model is not listed as an author. The mathematical claim rests on the explicit point set, certificate, and executable exact checkers.

Status. This is a working preprint. At the time of writing, the certificate has been reproduced with the included exact checkers but has not undergone independent peer review.

References

[1] E. Friedman, Packing unit squares in squares: A survey and new results, *Electron. J. Combin.*, Dynamic Survey DS7 (2009), first published 1998. `https://doi.org/10.37236/28`.

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[3] D. Ellsworth, Squares in squares, high-precision online packing catalogue, accessed 5 August 2026. `https://kingbird.myphotos.cc/packing/squares_in_squares.html`.

[4] S. Fort, Packing 17 unit squares: exact lower bound $s(17) > 4.456575$, computer-assisted certificate repository (2026). `https://github.com/stanislavfort/17squares`.